Card and Krueger (1994): minimum wage and employment

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Research question and motivation

This project studies whether New Jersey's April 1992 increase in its minimum wage reduced fast-food restaurant employment relative to eastern Pennsylvania. The paper's central policy question is whether the standard competitive-market prediction of lower employment is supported by the survey evidence.

Data and sample construction

The input is the course-provided data/raw/public.csv restaurant survey (410 stores; 331 in New Jersey and 79 in Pennsylvania). The first survey was conducted in February-March 1992 and the follow-up in November-December 1992. Full-time-equivalent employment is full-time workers plus managers plus one-half of part-time workers. The six permanently closed restaurants are assigned zero follow-up employment; temporary closures are missing in the main estimates and set to zero only in Table 3's designated sensitivity row.

The Table 4 regression sample contains 357 stores. It requires observed wave-1 and wave-2 FTE employment and a wave-1 starting wage; wave-2 wage is required for open/responding stores, while permanently closed stores are retained with follow-up employment set to zero. The treatment gap is (5.05 - wave-1 starting wage) / wave-1 starting wage for New Jersey stores initially below $5.05, and zero otherwise. Models use OLS conventional standard errors to match the paper's reported table.

Empirical design

The core design is a two-period difference-in-differences comparison between New Jersey (treated) and Pennsylvania (control). Table 3 reports state means, changes, a balanced sample, and New Jersey groups split by initial wage. Table 4 regresses the change in store FTE employment on a New Jersey indicator or the continuous initial wage gap, adding chain/ownership and regional controls.

Core replication results

The wave-1 mean FTE employment is 20.44 in New Jersey and 23.33 in Pennsylvania. The Table 3 change-in-means difference (New Jersey minus Pennsylvania) is 2.75 FTE employees per store.

Table 4 regression estimates

Model	Main coefficient	Standard error	N
(i) NJ dummy	2.33	(1.19)	357
(ii) NJ dummy	2.30	(1.20)	357
(iii) Initial wage gap	15.65	(6.08)	357
(iv) Initial wage gap	14.92	(6.21)	357
(v) Initial wage gap	11.98	(7.42)	357

The estimates are close to the paper's Table 4 benchmark. Small differences remain in the region-controlled wage-gap specification and may reflect the course package's coding, sample conventions, or source transcription. Table 2 and Table 3 outputs are generated from the same documented input and definitions.

Table 2: selected state means

Variable	New Jersey	Pennsylvania	NJ-PA t-stat
FTE employment, wave 1	20.44 (n=321)	23.33 (n=77)	-2.35
Full-time employees (%)	32.85 (n=321)	35.04 (n=77)	-0.73
Starting wage, wave 1 ($)	4.61 (n=314)	4.63 (n=76)	-0.41
Starting wage = $4.25 (%)	30.51 (n=331)	32.91 (n=79)	-0.41
Full meal price, wave 1 ($)	3.35 (n=311)	3.04 (n=76)	3.79
FTE employment, wave 2	21.03 (n=319)	21.17 (n=77)	-0.12
Starting wage, wave 2 ($)	5.08 (n=318)	4.62 (n=71)	19.71
Starting wage = $5.05, wave 2 (%)	85.50 (n=331)	1.27 (n=79)	20.96
Full meal price, wave 2 ($)	3.41 (n=305)	3.03 (n=71)	4.73

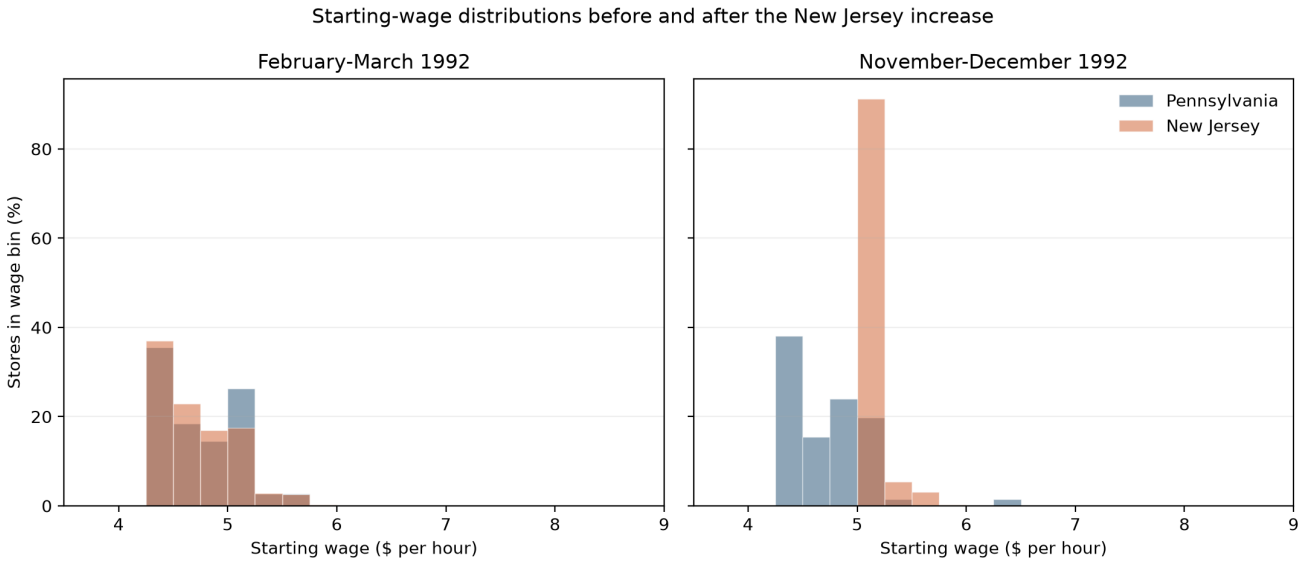
Category-share rows for starting wages use all sampled stores as the denominator; missing second-wave wages are not counted in either wage category. Table 2's full output, including store composition, standard errors, and tests, is in outputs/tables/table2_summary.csv.

Table 3: employment comparisons

Table 3 row	Comparison	Estimate (FTE)	SE	t-stat
1	New Jersey - Pennsylvania	-2.89	(1.44)	-2.00
2	New Jersey - Pennsylvania	-0.14	(1.08)	-0.13
3	New Jersey - Pennsylvania	2.75	(1.34)	2.05
4	New Jersey - Pennsylvania	2.75	(1.34)	2.05
5	New Jersey - Pennsylvania	2.51	(1.35)	1.86
3	NJ low wage - high wage	3.36	(1.30)	2.58
3	NJ midrange wage - high wage	2.91	(1.22)	2.39

The low-, middle-, and high-initial-wage New Jersey groups contain 101, 140, and 73 stores before outcome-specific missingness. Row 3's standard errors now use paired within-store employment changes for the stores observed in both waves; the earlier independent-wave formula overstated their uncertainty. The NJ-PA row-3 estimate is 2.75 (SE 1.34; t=2.05), close to the paper's text-reported 2.76 and t=2.03. Row 4 independently confirms the balanced-sample result: 2.75 (SE 1.34). Full subgroup means, sample sizes, standard errors, and tests are in outputs/tables/table3_employment.csv.

Figure 1



The figure shows the percentage distribution of observed starting wages by state in each survey wave. Wave-2 wages are not imputed for closures or nonrespondents.

Independent extension: ownership heterogeneity

As an independent extension, the project tests whether the NJ-policy employment effect differs between company-owned and franchise-owned restaurants. The two-wave store-level model includes store fixed effects, a wave indicator, and NJ-by-post interactions for ownership status; standard errors are clustered by store. The additional NJ-by-post interaction for company ownership is -1.04 (clustered SE 3.40, p=0.760). The estimated NJ-by-post effect among franchise-owned stores is 3.13. This is an exploratory heterogeneity test, not evidence that ownership status was randomly assigned.

Validation, limitations, and conclusion

The workflow checks the 410-record input, the expected 357-observation Table 4 sample, and the key published Table 4 coefficients. Table 2's means use variable-specific nonmissing observations. The main limitations are the nonrandom state-level treatment, one pre-treatment wave, a modest store sample, survey measurement error, and limited external validity beyond the sampled chains and regions. Differences in data coverage and closure handling are reported rather than hidden.